CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 2: Polynomials

Previous year board questions, topic-wise: 13 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Zeroes and coefficients of a quadratic polynomial

1. α and β are the zeroes of $p(x) = x^2 - 3x - 1$. Find $\frac{1}{\alpha} + \frac{1}{\beta}$. CBSE 2026, 2 marks

2. Assertion (A): The polynomial $p(y) = y^2 + 4y + 3$ has two zeroes. Reason (R): A quadratic polynomial has at most two zeroes. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2026, 1 mark

3. The zeroes α and β of the polynomial $3x^2 + 6x + k$ satisfy $\alpha + \beta + \alpha\beta = -\frac{2}{3}$. Find k. CBSE 2025, 1 mark

4. Find the zeroes of the polynomial $x^2 + \frac{4}{3}x - \frac{4}{3}$. CBSE 2025, 2 marks

5. The zeroes of $p(x) = 2x^2 - k\sqrt{2}\,x + 1$ add up to $\sqrt{2}$. Find k. (a) $\sqrt{2}$ (b) 2 (c) $2\sqrt{2}$ (d) $\frac{1}{2}$ CBSE 2024, 1 mark

6. Each zero of $x^2 + px + q$ is double a zero of $4x^2 - 5x - 6$. Find p. (a) $-\frac{5}{2}$ (b) $\frac{5}{2}$ (c) -5 (d) 10 CBSE 2024, 1 mark

7. α and β are the zeroes of $p(x) = x^2 + x - 1$. Find $\frac{1}{\alpha} + \frac{1}{\beta}$. (a) 1 (b) 2 (c) -1 (d) $-\frac{1}{2}$ CBSE 2023, 1 mark

8. Which of these is a quadratic polynomial with zeroes $-\frac{2}{3}$ and $\frac{2}{3}$? (a) $4x^2 - 9$ (b) $\frac{4}{9}(9x^2 + 4)$ (c) $x^2 + \frac{9}{4}$ (d) $5(9x^2 - 4)$ CBSE 2023, 1 mark

9. The sum of the zeroes of $x^2 - (k+6)x + 2(2k-1)$ is half of their product. Find k. CBSE 2019, 3 marks

2 Zeroes from the graph

10. The graph of $y = f(x)$ is a curve that stays above the x-axis everywhere and never touches it. How many zeroes does $f(x)$ have? (a) 0 (b) 1 (c) 2 (d) 4 CBSE 2026, 1 mark

11. The graphs of two quadratic polynomials are drawn on the same axes. One parabola opens upwards and the other opens downwards, and both cut the x-axis at the same two points. How many distinct zeroes do the two polynomials have between them? CBSE 2025, 1 mark

12. Assertion (A): a polynomial whose graph touches the x-axis at exactly one point cannot be quadratic. Reason (R): a polynomial of degree n, with $n > 1$, has at most n zeroes. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true. CBSE 2024, 1 mark

13. The graph of $y = p(x)$ is a parabola that opens downwards and touches the x-axis at exactly one point, without crossing it. How many zeroes does $p(x)$ have? (a) 3 (b) 1 (c) 2 (d) 0 CBSE 2023, 1 mark

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